CS & IT ENGINEERING

Computer Organization Architecture

Instruction and Addressing Modes

DPP 01 Discussion Notes

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[NAT]

#Q. A relative branch mode type instruction is stored in memory starting from address 240. The branch is made to an address 140. What should be the value of relative address field of the instruction, if each instruction is stored on 2 memory locations?

Note: All numbers are in decimal

240	Instn
242	Next instn

Target = 140 = PC + offset

140 = 242 + offset

offset = 140 − 242

= 98

[MCQ]

#Q. Consider the following:

1. Operation code
2. Source operand reference
3. Result operand reference
4. Next instruction reference

Which of the above are typical elements of machine instructions?

A. 1, 2 and 3 only

B. 1, 2 and 4 only

C. 3 and 4 only

D. 1, 2, 3 and 4

[MCQ]

#Q. Which addressing mode helps to access table data in memory efficiently?

→ array

A. Indirect mode

B. Immediate mode

C. Auto-increment or Auto-decrement mode ✓

D. Index mode

[MCQ]

ADD SUB MUL

#Q. An addressing mode in which the location of the data is contained within the mnemonic, is known as?

A. Immediate addressing mode

B. Implied addressing mode

C. Register addressing mode

D. Direct addressing mode

[MCQ]

#Q. The addressing modes used for source operand in the following instructions are respectively?

Reg. mode (destinat)

R1 ← #5

R1 ← M[5000]

R1 ← M[R2]

A. Implied, direct, register

B. Implied, direct, register indirect

C. Immediate, direct, register indirect

D. Immediate, direct, register

[NAT]

Ans = 556

#Q. Consider a PC-relative mode type branch instruction which takes branch on address 720 in memory. The instruction has offset value 160. What is the starting address of this instruction in memory, if each instruction is stored in memory on 4 locations?

Note: All numbers are in decimal

Target (E.A.) = 720

offset = 160

720 = PC Value + 160

PC = 720 − 160

= 560

[MCQ]

#Q. Consider the system in which in fetch cycle complete instruction is fetched. Which of the following addressing modes do(es) not require memory access for operand after fetch cycle?

A. Register Mode

B. Register Indirect Mode

C. Indirect Mode

D. Indexed Mode

[NAT]

Ans = 2

#Q. Consider a system which support only 2 address instructions only, and supports word addressable memory. The memory has total capacity of 2MB with word size of 4 Bytes. The system supports 350 distinct instructions. How many memory locations are required to store each instruction in the memory?

→ 32 bits

→ opcode = 9 bits

opcode	a.1	a.2
9	19	19

47-bits

no. of cells = $\frac{2MB}{4B}$

$= \frac{2^1 \cdot 2^{20}}{2^2} = 2^{19}$ ⇒ add. = 19 bits

[NAT]

Ans = 4000

#Q. Consider a system which support only 2 address instructions only, and supports word addressable memory. The memory has total capacity of 2MB with word size of 4 Bytes. The system supports 350 distinct instructions. If a program has 500 instructions, which is stored in the memory then the amount of memory required to store the entire program is ____ bytes?

for 1 instn ⇒ 2 locations

500 —||— ⇒ 2 * 500 = 1000 locations

= 1000 * 4B

= 4000 Bytes

[MCQ]

2^3 $2^{18} \Rightarrow$ add. = 18 bits

#Q. The word addressable memory of a computer has 256K words of 32-bit each. The computer has an instruction format with four fields; an operation code field, a mode field to specify one of 8 addressing modes, a register address field to specify one of the 64 processor registers and a memory address field.

Reg. = 6 bits

The bits for each field required in instruction format if the instruction is stored exactly in one word in memory?

A Opcode: 5, Addressing mode: 3, Register: 6, Memory address: 20

B Opcode: 3, Addressing mode: 3, Register: 6, Memory address: 20

C Opcode: 5, Addressing mode: 3, Register: 6, Memory address: 18

D Opcode: 3, Addressing mode: 3, Register: 6, Memory address: 18

32
opcode
mode
Reg.
Mem. address
3
6
18
5-bits

[NAT]

Ans = 64

opcode = 8 bits

#Q. A digital computer has a memory unit with 32-bits per word. The instruction set consists of 240 different operations. All the instructions have an operation code part (opcode) and an address part (allowed for only 1 address). Each instruction is stored in one word of memory. The maximum allowable size of memory (word addressable) is _____Mbytes?

no. of cells in mem. $= 2^{24}$

mem. size $= 2^{24} * 32$ bits

$= 2^{24} * 4\,B$

$= 2^{24} * 2^{2}\,B$

$= 2^{26}\,B$

$= 64\,MB$

[NAT]

Ans = 1024

#Q. Consider a system which supports 2-address and 1-address instructions. The system uses 16 bits instructions and 5-bits addresses. If there are total 32 2-address instructions then maximum how many 1-address instructions can be formulated?

2-add. instⁿ

1-add. instⁿ

max = $2^6 = 64$
used = 32
unused = 32

max = $32 * 2^5 = 2^{10} = 1024$

[NAT]

Ans = 508

#Q. Consider a system which supports 3-address, 2-address and 1 -address instructions. It has 32-bit instructions with 8-bits addresses. If there are 254 3-address instructions and <u>1024</u> 1-address instructions, then maximum how many 2-address instructions can be formulated?

3-add. instⁿ

max = 2^8 = 256
used = 254
unused = 2

2-add. instⁿ

max = $2 * 2^8$ = 512
used = x
unused = $(512 - x)$

1-add. instⁿ

max = $(512 - x) * 2^8$ = 1024

$(512 - x) = 2^2$

$x = 512 - 4 = \underline{508}$ Ans.

[NAT]

Ans = 128

#Q. Consider a system which supports 2-address and 1-address instructions. The system has 18 bits instructions. If there are 7 2-address instructions and 1152 1-address instructions, then the maximum size of memory supported by system is ____bytes?

2-add. instn — 18-bits

opcode	a.1	a.2
4	7	7

14 bits

max = $2^4 = 16$

used = 7

unused = 9

1-add. instn — 18

opcode	a.1
11	7

4 | 7

max = $9 * 2^7 = 1152$

No. of cells = 2^7

mem. size

$= 2^7 * 1B$

$= 128B$

[MCQ]

#Q. Consider a system which supports 2-address, 1-address and 0-address instructions. The system has 'i' bits instructions and 'a' bits addresses. If there are 'x' 2-address instructions and 'y' 1-address instructions then which of the following is correct for maximum number of 0-address instructions supported by system?

A. $2^i - 2^a x - y$

B. $2^i - 2^{2a} x - y$

C. $2^i - 2^{2a} x - y2^a$

D. $2^i - 2^a x - y2^a$

2-add. ← i →

Opcode	a.1	a.2
$i-2a$	a	a

$$\max = 2^{i-2a}$$
$$\text{used} = x$$
$$\text{unused} = 2^{i-2a} - x$$

1-add.

$$max = \left(2^{i-2a} - x\right) * 2^{a}$$

$$= 2^{i-a} - x2^{a}$$

$$used = y$$

$$unused = 2^{i-a} - x2^{a} - y$$

0- add.

$$max = \left(2^{i-a} - x2^{a} - y\right) * 2^{a}$$

$$= 2^{i} - x2^{2a} - y2^{a}$$

[NAT]

Ans = 19

#Q. Consider there are 4 types of instructions in system:

Type 1: One opcode and 2 registers

Type 2: One opcode and 1 register

Type 3: One opcode and 1 memory address

Type 4: One opcode, 1 register and 1 memory address

Number of registers in CPU=128 $= 2^7 \Rightarrow$ Reg. = 7 bits

Maximum instruction length: 32bits (Variable length instructions supported)

Total Instructions: Type-1: 15, Type-2: 20, Type-3: 12, Type-4: 14

Maximum memory address size = 19 bits

Total instns

$= 15 + 20 + 12 + 14$

$= 61$

$\Downarrow$

opcode = 6-bits

Type 1

Type 2

Type 3

Type 4

[NAT]

Ans = 3

#Q. Consider a register-based architecture system which can support maximum 2-address instructions. For this system the following intermediate code is going to be converted in machine code. Minimum how many registers are required in system so that the code can run without register spill?

t1 = X + Y
t2 = Z * 2
t3 = t2 + A
t4 = t3 – t1
t5 = t4 + t3

R1 ← X
R2 ← Y
R1 ← R1 + R2
R2 ← Z
R2 ← R2 * 2
R3 ← A
R2 ← R2 + R3
R3 ← R2
R2 ← R2 – R1
R2 ← R2 + R3

Note: X, Y, A and Z are memory operands; and consider first operand as destination operand and there is no any optimization done by compiler.

[NAT]

#Q. Consider a register-memory based architecture system which can support maximum 2-address instructions. For this system the following intermediate code is going to be converted in machine code. Minimum how many registers are required in system so that the code can run without register spill?

t1 = X + Y
t2 = Z * X
t3 = t2 + t1
t4 = t3 – Y
t5 = t4 + t3

2.

Note: X, Y, and Z are memory operands; and consider first operand as destination operand and there is no any optimization done by compiler.

$R_1 \leftarrow X$

$R_1 \leftarrow R_1 + Y$

$R_2 \leftarrow Z$

$R_2 \leftarrow R_2 * X$

$R_2 \leftarrow R_2 + R_1$

$R_1 \leftarrow R_2$

$R_2 \leftarrow R_2 - Y$

$R_2 \leftarrow R_2 + R_1$

Mini. 2 registers required.

[NAT]

#Q. Consider a single AC-based architecture system which can support maximum 1-address instructions. For this system the following intermediate code is going to be converted in machine code. Minimum how many general-purpose registers are required in system so that the code can run without register spill?

t1 = X + Y

t2 = t1 * X

t3 = Y + Z

t4 = t3 – t2

t5 = t3 – t4

Note: X, Y, and Z are memory operands; and consider first operand as destination operand and there is no any optimization done by compiler.

2

$AC \leftarrow X$

$AC \leftarrow AC + Y$

$AC \leftarrow AC * X$

$R_1 \leftarrow AC$

$AC \leftarrow Y$

$AC \leftarrow AC + Z$

$R_2 \leftarrow AC$

$AC \leftarrow AC - R_1$

$R_1 \leftarrow AC$

$AC \leftarrow R_2$

$AC \leftarrow AC - R_1$

Registers required = 2.

THANK - YOU